1-(MAT/11_HL_Summer_2013_Q12)-*Quadratics,Graphs,Integration*

(a) Express $4x^2 - 4x + 5$ in the form $a(x-h)^2 + k$ where $a, h, k \in \mathbb{Q}$. *[2 marks]*

(b) The graph of $y = x^2$ is transformed onto the graph of $y = 4x^2 - 4x + 5$. Describe a sequence of transformations that does this, making the order of transformations clear. *[3 marks]*

The function f is defined by $f(x) = \dfrac{1}{4x^2 - 4x + 5}$.

(c) Sketch the graph of $y = f(x)$. *[2 marks]*

(d) Find the range of f. *[2 marks]*

(e) By using a suitable substitution show that $\int f(x)\,\mathrm{d}x = \dfrac{1}{4}\int \dfrac{1}{u^2+1}\,\mathrm{d}u$. *[3 marks]*

(f) Prove that $\displaystyle\int_1^{3.5} \frac{1}{4x^2 - 4x + 5}\,\mathrm{d}x = \frac{\pi}{16}$. *[7 marks]*

2-(MAT/12_HL_Summer_2014_Q4)-*Quadratics*

The roots of a quadratic equation $2x^2 + 4x - 1 = 0$ are α and β.

Without solving the equation,

(a) find the value of $\alpha^2 + \beta^2$; *[4]*

(b) find a quadratic equation with roots α^2 and β^2. *[2]*

3-(MAT/10_HL_Winter_2014_Q2)**-***Quadratics*

The quadratic equation $2x^2 - 8x + 1 = 0$ has roots α and β.

(a) Without solving the equation, find the value of

(i) $\alpha + \beta$;

(ii) $\alpha\beta$. *[2]*

Another quadratic equation $x^2 + px + q = 0$, $p, q \in \mathbb{Z}$ has roots $\dfrac{2}{\alpha}$ and $\dfrac{2}{\beta}$.

(b) Find the value of p and the value of q. *[4]*

4-(MAT/10_HL_Winter_2020_Q6)**-***Quadratics,Trigonometry*

[Maximum mark: 4]

Consider the equation $ax^2 + bx + c = 0$, where $a \neq 0$. Given that the roots of this equation are $x = \sin\theta$ and $x = \cos\theta$, show that $b^2 = a^2 + 2ac$.

5-(MathAA/11_SL_Summer_2021_Q7)**-***Quadratics*

[Maximum mark: 14]

Let $f(x) = mx^2 - 2mx$, where $x \in \mathbb{R}$ and $m \in \mathbb{R}$. The line $y = mx - 9$ meets the graph of f at exactly one point.

(a) Show that $m = 4$. [6]

The function f can be expressed in the form $f(x) = 4(x - p)(x - q)$, where p, $q \in \mathbb{R}$.

(b) Find the value of p and the value of q. [2]

The function f can also be expressed in the form $f(x) = 4(x - h)^2 - k$, where h, $k \in \mathbb{R}$.

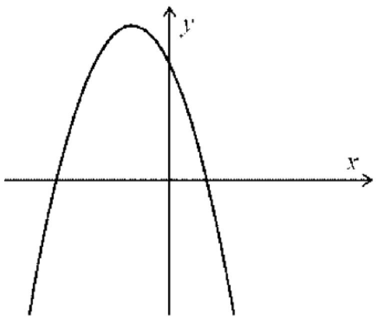
(c) Find the value of h and the value of k. [3]

(d) Hence find the values of x where the graph of f is both negative and increasing. [3]

6-(MathAA/10_SL_Winter_2021_Q1)-*Quadratics*

[Maximum mark: 7]

Consider the function $f(x) = -2(x-1)(x+3)$, for $x \in \mathbb{R}$. The following diagram shows part of the graph of f.



(a) For the graph of f

(i) find the x-coordinates of the x-intercepts;

(ii) find the coordinates of the vertex. [5]

The function f can be written in the form $f(x) = -2(x-h)^2 + k$.

(b) Write down the value of h and the value of k. [2]

7-(MathAA/10_HL_Winter_2021_Q7)-*Quadratics*

[Maximum mark: 7]

The equation $3px^2 + 2px + 1 - p$ has two real, distinct roots.

(a) Find the possible values for p. [5]

(b) Consider the case when $p = 4$. The roots of the equation can be expressed in the form $x = \frac{a \pm \sqrt{13}}{6}$, where $a \in \mathbb{Z}$. Find the value of a. [2]

1–(MAT/20_HL_Winter_2013_Q3)–*Graphs,Exponentials-Logarithms*

Consider $f(x) = \ln x - e^{\cos x}$, $0 < x \leq 10$.

(a) Sketch the graph of $y = f(x)$, stating the coordinates of any maximum and minimum points and points of intersection with the x-axis. *[5]*

(b) Solve the inequality $\ln x \leq e^{\cos x}$, $0 < x \leq 10$. *[2]*